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Abstract

Development of a Hybrid Robotic Surface Finishing Framework Based on Geometric Planning, Reinforcement Learning, Force-Aware Imitation Learning, and Feature Masking

This thesis draft proposes a hybrid robotic surface-finishing framework for processing an entire product rather than a single isolated surface region. The proposed framework separates the problem into four interacting modules. First, a tool-footprint-aware geometric planner generates a full-coverage path over a CAD or reconstructed surface while accounting for the effective polishing-tool width, overlap, surface boundary, tool orientation, and manipulator feasibility. Second, a reinforcement-learning policy modulates the sliding velocity along the reference path to reduce spatial variation of the pressure-velocity product and the resulting material removal. Third, a force-aware imitation-learning policy generates local pose-force action chunks for edges, narrow transitions, and hardware-adjacent regions that are difficult to represent through a single geometric rule set. Fourth, a feature-masking module uses object-centric region representations inspired by VIOLA and predictive future-mask plans inspired by Mask2Act to reduce reliance on reflections and background distractors.

All high-level trajectory generators share a common 9-D waypoint representation consisting of Cartesian position, orientation, and desired force. A deterministic low-level admittance controller executes the generated trajectory and compensates for measured force error. The proposed reinforcement-learning policy does not directly command robot joints or change the geometric path;

it produces only a bounded path-progress or velocity scale. Similarly, imitation learning is applied selectively to short difficult segments instead of the complete product surface, thereby reducing demonstration cost and preserving the interpretability of global coverage planning.

A guitar is selected as the primary demonstration object because a single instrument contains broad coated surfaces, strongly curved boundaries, cutaways, neck transitions, frets, pickup covers, bridges, and other small reflective metal components. The body provides a suitable test of CAD-based full coverage, whereas local components expose the perception and contact challenges targeted by force-aware imitation learning and feature masking. Planned experiments compare geometry-only planning, advanced contact-point planning, manual local programming, vision-only imitation learning, force-observation imitation learning, direct pose-force imitation learning, fixed speed, inverse-Preston control, model-based speed control, and reinforcement-learning-based velocity modulation. The main metrics are coverage ratio, uncovered boundary area, removal-rate and measured-removal variation, force tracking, contact loss, collision and intervention rates, programming and teaching time, and generalization to unseen geometry, appearance, wear, friction, and pad conditions.

Keywords: Robotic Surface Finishing, Full-Coverage Path Planning, Tool Footprint, Reinforcement Learning, Force-Aware Imitation Learning, Feature Masking, VIOLA, Mask2Act, Admittance Control

Chapter 1

Introduction

1.1 Background and Motivation

Robotic surface finishing, including polishing, sanding, grinding, and buffing, is a physically demanding and quality-sensitive manufacturing process. The final result depends not only on the nominal path of the robot but also on contact pressure, sliding velocity, dwell time, tool orientation, abrasive state, and local surface geometry. Conventional fixed-path automation can achieve high repeatability on structured parts, yet it is difficult to transfer to high-mix products containing large smooth regions, sharp boundaries, concave transitions, narrow features, and reflective hardware.

The dominant automation pipeline combines three-dimensional perception or CAD with explicit coverage planning and compliant control. A surface model provides normals, curvature, boundaries, and collision geometry; a planner generates a raster, iso-parametric, geodesic, or region-based trajectory; and a force, impedance, or admittance controller maintains contact. Recent work has improved uniform coverage, segmentation, contact-point switching, and manipulator feasibility [1–5]. These methods are effective on broad and continuously observable surfaces because product variation can be represented by a new CAD or mesh while reusing the same planning algorithm.

However, full-product polishing exposes two unresolved limitations. First, a path that covers a mathematical surface does not necessarily cover the surface with the physical tool. A polishing pad has a finite and deformable footprint. The path spacing must therefore depend on the effective tool width, overlap

ratio, surface curvature, tool tilt, and boundary offset. The tool center point (TCP) is a point and has no width; throughout this thesis, the phrase *tool width* refers to the effective contact footprint of the polishing tool. Second, geometric coverage does not guarantee uniform material removal because the actual pressure and sliding velocity vary during execution. According to the Preston relation, the local removal rate is approximately proportional to their product [6].

A third challenge occurs at edges, cutaways, narrow transitions, and hardware-adjacent regions. Modern geometric planners can introduce region-specific strategies and move the tool contact point, but their rule and parameter complexity grows with the number of local contact modes [1]. In repair and reconditioning, the geometry is further combined with product-specific scratches, oxidation, loss of gloss, and wear. The desired approach direction, tool tilt, contact-force reduction, repeated short strokes, and termination time may be easier for a skilled operator to demonstrate than to specify through a universal analytical objective.

This thesis therefore proposes a hybrid architecture. Tool-footprint-aware full-coverage planning is used for the entire CAD surface and broad regular regions. Reinforcement learning optimizes the sliding velocity along the geometric path while the path and low-level admittance controller remain fixed. Force-aware imitation learning generates local pose–force trajectories for difficult segments. Finally, object-centric and predictive feature masking is introduced to reduce the influence of reflections, background clutter, and irrelevant image regions on the imitation-learning policy.

Hybrid Full-Product Surface-Finishing Framework

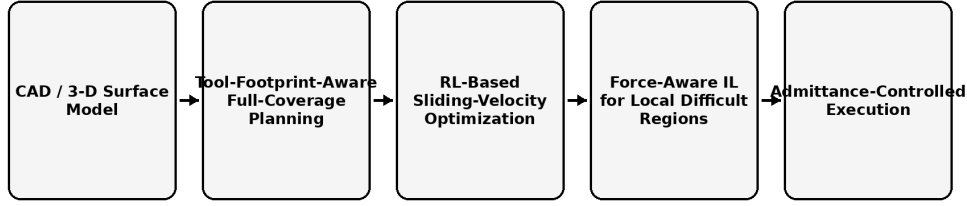


Figure 1.1: Overall structure of the proposed hybrid full-product surface-finishing framework.

1.2 Related Works

1.2.1 Surface Reconstruction and Coverage Planning

Scan-based robotic surface finishing has progressed from planar projection toward free-form surface reconstruction, segmentation, geodesic planning, and kinematics-aware optimization. Tafuro et al. studied planning from realistic industrial scans and emphasized incomplete reconstruction and abrupt curvature changes [2]. Schneyer et al. segmented free-form surfaces and selected different finishing-disk contact points to improve access to concave regions [1]. Gong et al. generated uniform coverage trajectories on equation-free mesh surfaces [3], while recent planners exploit redundancy, region decomposition, and task-aware partitioning [4, 5, 7].

These studies show that local feature coverage can be addressed geometrically when the workpiece and tool model are accurate. Nevertheless, every new contact mode may require a new region classifier, contact-point rule, clearance threshold, or trajectory topology. This thesis retains advanced geometric planning as a strong baseline but uses imitation learning where the desired strategy depends jointly on geometry, visual condition, and contact feedback.

1.2.2 Material-Removal Optimization and Reinforcement Learning

Coverage and removal uniformity are distinct. Recent polishing studies jointly consider force, position, and speed [8], constant-pressure processing on curved surfaces [9], or data-driven material-removal estimation [10]. Reinforcement learning has also been used to learn abrasive shape transitions or force-responsive finishing skills [11, 12]. In contrast, the proposed RL policy controls only a scalar path-progress rate above a fixed CAD path and admittance controller. This restricted action space is intended to improve safety, interpretability, and sim-to-real transfer.

1.2.3 Force-Aware Imitation Learning

Force-aware imitation learning has rapidly developed for contact-rich tasks. ForceMimic records force-motion demonstrations and predicts hybrid force-position parameters [13]. FoAR fuses RGB and high-frequency force/torque signals with a future-contact predictor [14]. DexForce computes force-informed actions from kinesthetic demonstrations [15], and Flow with the Force Field learns compliant flow-matching policies from demonstration-guided simulation data [16]. These methods establish that force is valuable in contact-rich policy learning, but they also prevent a broad novelty claim based only on force input or output. The contribution of this thesis is instead defined by selective local teaching, a continuous 9-D pose-force action chunk, direct integration with an admittance controller, and full-product coordination with global planning and RL process optimization.

1.2.4 Object-Centric and Mask-Guided Policy Learning

VIOLA constructs object-centric representations from general object proposals generated by a pretrained vision model and uses a transformer policy

to attend to task-relevant visual factors [17]. Mask2Act predicts future masks of task-relevant objects as latent plans from action-free video and uses them to guide policy learning with limited robot action data [18]. Both methods are relevant to polishing, where bright reflections, background texture, and non-target hardware can dominate a conventional image encoder. Chapter 6 adapts their ideas to a force-aware polishing policy.

1.3 Research Objectives

The objective is to develop and evaluate a modular framework with the following functions:

1. generate a full-coverage path over an entire CAD surface using the effective polishing-tool footprint and overlap ratio;
2. learn an adaptive sliding-velocity policy that reduces spatial variation of the Preston rate without directly controlling robot joints or the 6-D pose;
3. learn local pose–force action chunks from expert demonstrations for edges and contact-sensitive regions;
4. incorporate VIOLA-inspired region tokens and Mask2Act-inspired predictive mask plans into the force-aware IL framework;
5. validate the complete framework on a guitar-scale object using coverage, removal, force, safety, efficiency, and generalization metrics.

1.4 Thesis Organization

Chapter 2 defines the shared system interfaces and problem formulation. Chapter 3 presents tool-footprint-aware full-coverage path planning. Chapter 4

describes RL-based sliding-velocity optimization. Chapter 5 presents force-aware imitation learning. Chapter 6 proposes the feature-masking architecture. Chapter 7 defines the experimental design, and Chapter 8 concludes the thesis and outlines future work.

Chapter 2

System Overview and Problem Formulation

2.1 System Overview

The system consists of a six-degree-of-freedom manipulator, a polishing spindle, a wrist six-axis force/torque sensor, a CAD or reconstructed workpiece model, one or more RGB or RGB-D cameras, and a rigid fixture. The framework contains four high-level components: a geometric full-coverage planner, an RL process optimizer, a force-aware IL local planner, and a feature-masking module. All trajectories are executed by a deterministic low-level admittance controller and filtered by an independent safety supervisor.

2.2 Task Decomposition

Let $\mathcal{S}$ denote the workpiece surface and $\Omega \subset \mathcal{S}$ the target finishing region. The complete task is partitioned into a global geometric region Ω_g and a set of local difficult regions $\{\Omega_\ell^i\}_{i=1}^{N_\ell}$:

$$\Omega = \Omega_g \cup \bigcup_{i=1}^{N_\ell} \Omega_\ell^i, \quad \Omega_g \cap \Omega_\ell^i = \emptyset. \quad (2.1)$$

The global region contains broad, continuous, and reliably modeled surfaces. Local regions contain sharp boundaries, small reflective hardware, narrow clearance, incomplete reconstruction, or contact behavior that requires configuration-dependent expert adjustment. Initial experiments may use manual region annotation; automatic assignment based on reconstruction confidence, curvature, clearance, and planner feasibility is reserved for future

System Overview and Shared Interfaces

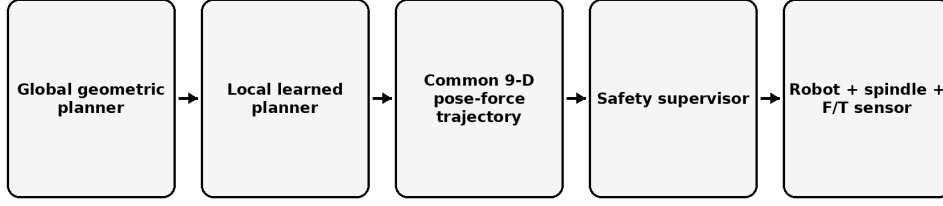


Figure 2.1: System modules and shared interfaces.

work.

2.3 Common Pose–Force Trajectory

Both the geometric planner and the IL policy generate a dense sequence of 9-D waypoints:

$$\mathbf{a}_k = [x_k, y_k, z_k, \omega_{x,k}, \omega_{y,k}, \omega_{z,k}, F_{x,k}, F_{y,k}, F_{z,k}]^\top. \quad (2.2)$$

The first six variables define the desired end-effector pose, with orientation represented as a rotation vector. The final three variables define the desired Cartesian force. The global planner derives pose from CAD geometry and uses a prescribed force profile. The local IL policy predicts the complete pose–force action chunk from sensory observations.

2.4 Admittance-Controlled Execution

The high-level trajectory does not directly produce joint torques. A Cartesian admittance controller computes a compliant pose correction from the desired and measured wrench:

$$\mathbf{M}_d \Delta \ddot{\mathbf{x}} + \mathbf{D}_d \Delta \dot{\mathbf{x}} + \mathbf{K}_d \Delta \mathbf{x} = \mathbf{F}_d - \mathbf{F}_m. \quad (2.3)$$

The controller maintains high-rate contact stability, while the high-level modules determine the global path, local strategy, and desired sliding speed. This separation is essential for fair evaluation: improvements caused by deterministic force control must not be incorrectly attributed to a learning policy.

2.5 Safety Supervisor

Hard limits are imposed on workspace, joint range, self-collision, workpiece collision, force, torque, velocity, acceleration, spindle state, and cumulative dwell. The supervisor can clip a velocity scale, reject a predicted waypoint, pause the spindle, or command a controlled retraction. Learning policies are therefore optimized within a bounded execution envelope.

2.6 Demonstration Object: Guitar

A guitar is used as the principal validation object because it combines broad coated surfaces and local difficult features in one product. The body top, back, and side are suitable for CAD-based coverage. The body edge, cutaway, neck joint, frets, pickup covers, bridge, and other metal hardware expose narrow, reflective, and contact-sensitive regions. The object is also compact enough for a UR10e-class robot and supports both manufacturing-oriented body finishing and repair-oriented local reconditioning.

Chapter 3

Tool-Footprint-Aware Full-Coverage Path Planning on CAD Surfaces

3.1 Problem Definition

The objective of this chapter is to generate a robot-executable path whose swept polishing footprint covers an entire target region on a CAD surface. The TCP is a point and does not possess a physical width. Therefore, the planner uses the effective polishing-tool footprint, denoted by w_{eff} , which depends on pad diameter, compression, tool tilt, local curvature, and the selected contact point.

Let $\gamma(s) \in \mathcal{S}$ denote the centerline of the polishing path, parameterized by arc length s , and let $\mathcal{B}(\gamma(s), \mathbf{R}(s), w_{\text{eff}})$ denote the surface footprint covered by the tool at that configuration. Full coverage is achieved when

$$\Omega \subseteq \bigcup_{s \in [0, L]} \mathcal{B}(\gamma(s), \mathbf{R}(s), w_{\text{eff}}). \quad (3.1)$$

In practice, exact full coverage is relaxed by a small tolerance near protected boundaries and physically inaccessible regions.

3.2 CAD Surface Preprocessing

The CAD model is converted into a triangular mesh with controlled resolution. The target region is selected through CAD semantics, manual annotation, or surface segmentation. Boundary loops, normals, principal curvatures, and adjacency are computed. Mesh quality is checked for holes,

Tool-Footprint-Aware Full-Coverage Planning

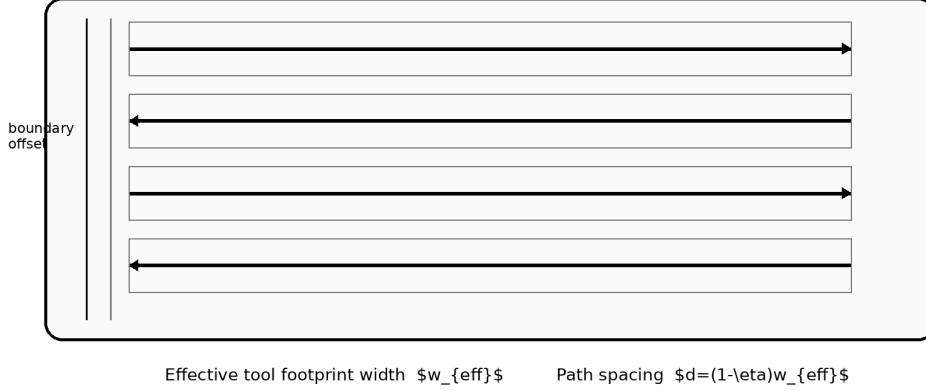


Figure 3.1: Concept of tool-footprint-aware full coverage. The path spacing is determined by the effective contact width and the desired overlap.

non-manifold edges, and inconsistent normals. If the product pose is obtained from a scan, scan-to-CAD registration determines the transformation to the robot base frame.

3.3 Effective Tool-Footprint Model

For a nominal circular pad with diameter D , the footprint projected onto the local tangent plane is approximately circular under normal contact and elliptical under tilt. A first-order effective cross-track width is

$$w_{\text{eff}}(\mathbf{q}, \theta) = D g_c(\kappa_1, \kappa_2) g_t(\theta) g_p(F_n), \quad (3.2)$$

where g_c represents curvature-induced footprint change, g_t represents tool-tilt projection, and g_p represents compression under normal force. These factors can initially be calibrated empirically and later estimated online.

Given a desired overlap ratio $\eta \in [0, 1)$, the nominal path spacing is

$$d(\mathbf{q}) = (1 - \eta)w_{\text{eff}}(\mathbf{q}). \quad (3.3)$$

The spacing is reduced near high curvature, uncertain reconstruction, or regions where the footprint projection changes rapidly.

3.4 Coverage Path Generation

The planner supports two complementary methods. The first generates scan lines on a reference parameterization and projects them onto the CAD mesh. The second creates geodesic offset curves from a selected seed curve or boundary. In both cases, the cross-track distance is adapted according to $d(\mathbf{q})$ rather than a fixed Euclidean spacing.

Candidate path directions are evaluated using coverage efficiency, number of disconnected segments, curvature variation, path length, reachability, and boundary behavior. The selected centerlines are ordered in a serpentine sequence and connected through safe transition trajectories. Geodesic Hermite splines or another surface-constrained smoother reduce abrupt direction changes without leaving the CAD surface.

3.5 Boundary and Edge Treatment

For full footprint support, the centerline should remain approximately $w_{\text{eff}}/2$ inside the target boundary. This creates a trade-off: a conservative offset prevents pad overhang but can leave an uncovered strip. The planner therefore offers three boundary modes:

1. **Conservative offset:** the complete footprint remains inside the target region;
2. **Controlled overhang:** partial footprint overhang is allowed when surrounding geometry is safe;
3. **Local strategy handoff:** the boundary strip is delegated to the force-aware IL policy or a specialized contact-point planner.

The third mode is central to the hybrid framework. It prevents the global planner from accumulating a large number of product-specific edge rules.

3.6 Tool Pose and Robot Feasibility

Tool orientation is initialized from the local normal $\mathbf{n}(\mathbf{q})$ and path tangent $\mathbf{t}(\mathbf{q})$. Orientation smoothing minimizes rotational acceleration while preserving a bounded angle to the surface normal. Inverse kinematics, joint limits, singularity measures, self-collision, fixture collision, and tool-body collision are evaluated at each waypoint. Infeasible segments are replanned or assigned to the local policy.

3.7 Optimization Objective

The global path is selected by minimizing

$$J = w_{\text{cov}}E_{\text{cov}} + w_{\text{ov}}E_{\text{overlap}} + w_L L + w_R E_{\text{rot}} + w_M E_{\text{manip}} + w_C E_{\text{collision}}, \quad (3.4)$$

where E_{cov} measures uncovered area, E_{overlap} penalizes excessive repeated processing, L is path length, E_{rot} measures orientation discontinuity, and the remaining terms penalize poor robot feasibility. Material-removal uniformity is not included as a purely geometric term; it is addressed online by the RL module in Chapter 4.

3.8 Output Trajectory

The optimized surface path is interpolated into the common 9-D pose–force format. Position and orientation follow the CAD geometry. The desired force is initially a constant normal-force profile or a process-specific schedule. Sliding speed is not fixed permanently; a nominal speed is stored and modulated online by the RL policy.

Chapter 4

Reinforcement Learning for Adaptive Sliding-Velocity Optimization

4.1 Motivation

A full-coverage path guarantees that the tool footprint visits the target surface, but it does not guarantee uniform material removal. Contact force varies with curvature, fixture error, pad deformation, sensor bias, and admittance dynamics. Under a constant feed speed, these force variations produce non-uniform pressure-velocity products and therefore non-uniform removal.

4.2 Restricted RL Formulation

The policy does not generate robot joint commands or a new Cartesian pose. At time t , the observation is

$$\mathbf{o}_t^{\text{RL}} = [F_{n,t}, \mathbf{F}_{t-h:t}, F_n^*, \dot{s}_t, \kappa_t, \phi_t, \mathbf{x}_{a,t}], \quad (4.1)$$

where $F_{n,t}$ is the measured normal force, $\mathbf{F}_{t-h:t}$ is a wrench history, F_n^* is the desired force, $\dot{s}_t$ is current path progress, κ_t is local curvature, ϕ_t is the normalized path phase, and $\mathbf{x}_{a,t}$ contains admittance states.

The action is a bounded velocity scale

$$\alpha_t \in [\alpha_{\min}, \alpha_{\max}], \quad \dot{s}_{t+1} = \alpha_t \dot{s}_{\text{nom}}. \quad (4.2)$$

Action-rate limits and low-pass filtering prevent chatter.

RL-Based Adaptive Sliding-Velocity Optimization

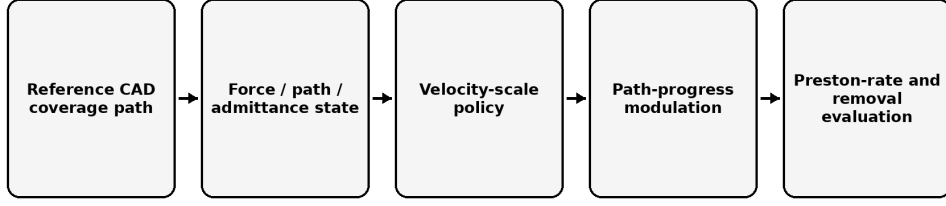


Figure 4.1: Hierarchical RL formulation. The policy changes only the path-progress rate while preserving the CAD path and the low-level admittance controller.

4.3 Preston-Based Reward

The local material-removal rate is approximated by

$$\rho_t = K p_t v_t, \quad (4.3)$$

where p_t is estimated from normal force and contact area and v_t is the sliding velocity. The policy tracks a target rate ρ^* :

$$r_t = -w_\rho(\rho_t - \rho^*)^2 - w_\alpha(\Delta\alpha_t)^2 - w_J J_t^2 - w_S P_{\text{safety}}. \quad (4.4)$$

The smoothness term penalizes rapid velocity changes, and P_{safety} penalizes overload, contact loss, excessive tracking error, or unsafe tool motion.

4.4 Training Environment

Training is performed in a parallel simulation environment using the same reference paths used by the real controller. Domain randomization covers curvature, friction, pad stiffness, force-sensor bias, delay, noise, initial pose error, and contact parameters. The policy can be trained using PPO or another actor-critic method, although the exact algorithm is secondary to the restricted hierarchical formulation.

4.5 Analytical and Model-Based Baselines

A strong inverse-Preston controller is

$$v_t = \text{clip} \left(\frac{c}{p_t + \epsilon}, v_{\min}, v_{\max} \right). \quad (4.5)$$

A filtered PID or model-predictive speed controller can regulate $\rho_t - \rho^*$. These baselines are essential. RL is justified only if it provides better robustness under delay, saturation, curvature transitions, contact-area uncertainty, and noisy force estimates. If the analytical controller performs equivalently, RL adds unnecessary complexity.

4.6 Internal Preliminary Results

Internal simulation work has shown that adaptive velocity can strongly reduce Preston-rate variation. On a flat surface, the coefficient of variation decreased from 0.101 under fixed velocity to 0.013 under the learned policy. On a convex surface, the value decreased from 0.195 to 0.026. These results are preliminary and should not be interpreted as measured material-removal performance. Real-robot experiments and direct removal measurements are required.

4.7 Sim-to-Real Transfer and Safety

The real system uses the same observation normalization, action limits, and path-phase representation as simulation. Safety overrides the learned scale when force or path error exceeds a threshold. A gradual deployment sequence is used: offline replay, zero-contact motion, compliant contact on sacrificial coupons, and finally workpiece experiments. The learned policy is never allowed to modify the geometric path or exceed the certified speed and force envelope.

Chapter 5

Force-Aware Imitation Learning for Local Difficult Regions

5.1 Motivation and Scope

The local policy is designed for short segments that are difficult to cover reliably with a single global geometric strategy. Representative regions include body edges, cutaways, neck joints, fret areas, pickup covers, bridges, and narrow hardware clearances. The method is not intended to replace the CAD planner on broad surfaces, where explicit coverage and removal objectives are easier to verify.

Imitation learning is selected instead of unrestricted RL because a skilled operator can directly demonstrate the required approach direction, tool tilt, force reduction, repeated stroke, and termination. Real-world trial-and-error is expensive and potentially destructive, and a complete local-finishing reward would require difficult weighting among coverage, collision, force, surface damage, and stopping.

5.2 Demonstration Data

Each demonstration contains synchronized local RGB observations, optional masks, robot or demonstrator pose, six-axis wrench, spindle state, region identity, and termination labels. Demonstrations can be collected through force-feedback teleoperation, kinesthetic teaching with a passive tool, or a force-motion capture device. The dataset is segmented into approach, contact, processing, and retraction phases.

Force-Aware Imitation Learning for Local Difficult Regions

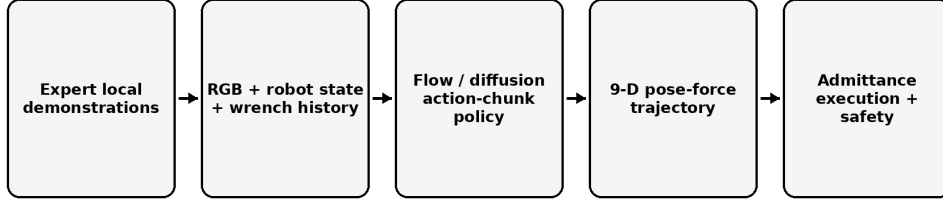


Figure 5.1: Force-aware imitation-learning pipeline for local difficult regions.

Force signals are bias-corrected, transformed to the tool frame, and filtered without eliminating contact onset or impact transients. Failed demonstrations are rejected or labeled explicitly. The local nature of the task reduces the duration and quantity of demonstrations compared with teaching the entire product surface.

5.3 Observation and Action

Before the feature-masking extension of Chapter 6, the baseline observation is

$$\mathbf{o}_t^{\text{IL}} = \{I_t, \mathbf{s}_t, \mathbf{H}_t^w\}, \quad (5.1)$$

where I_t is the RGB image, $\mathbf{s}_t$ is the robot state, and $\mathbf{H}_t^w$ is the recent wrench history. The action is an H -step chunk of 9-D pose-force waypoints:

$$\mathbf{A}_t = [\mathbf{a}_t, \mathbf{a}_{t+1}, \dots, \mathbf{a}_{t+H-1}] \in \mathbb{R}^{H \times 9}. \quad (5.2)$$

Only the first part of the chunk is executed before the policy replans from the latest observation.

5.4 Policy Architecture

A visual encoder extracts image features, a temporal encoder processes the wrench history, and a state encoder processes pose and velocity. The modalities are fused and passed to a conditional flow-matching or diffusion action generator. A generative action-chunk policy is preferred over one-step regression because multiple safe local trajectories may exist around the same hardware geometry.

5.5 Direct Desired-Force Generation

The policy explicitly predicts the desired Cartesian force together with motion. This differs from force-as-observation-only policies and from methods that convert force into a virtual positional target. Nevertheless, ForceMimic predicts wrench-position parameters and executes them using hybrid force-position control [13], while Min et al. separate desired motion and desired force for complex wooden-surface finishing [19]. The thesis therefore treats direct force output as one component of the contribution, not as the complete novelty claim.

5.6 Low-Level Execution and Safety

The desired pose–force chunk is passed to the common admittance controller. The safety supervisor validates workspace, collision margin, force, torque, velocity, spindle state, and cumulative dwell. Out-of-distribution detection can use latent feature distance, ensemble disagreement, or a task-specific confidence model. Low confidence triggers a stop or operator request rather than autonomous extrapolation.

5.7 Baselines and Ablations

The local policy is compared with a manually programmed path, RGB-only IL with fixed force, RGB-plus-wrench IL with fixed force, and direct 9-D pose-force IL. Ablations include wrench history length, action horizon, force observation, direct force output, flow matching versus deterministic behavior cloning, and task-relevant masking.

Chapter 6

Feature Masking for Object-Centric and Predictive Visual Representation

6.1 Motivation

Reflective guitar finishes and metal hardware generate highlights that can move with illumination and camera pose. A conventional RGB encoder may attend to these correlations rather than the target surface, tool contact region, defect, or protected obstacle. Background clutter and robot-body pixels can also dominate the learned representation. This chapter proposes a mask-guided extension of the force-aware IL framework. The design is preliminary and is presented as a research plan rather than a completed method.

6.2 Task-Relevant Mask Definitions

The proposed system represents four semantic mask types:

1. **Target-processing mask** M^{tar} : the region that should be polished;
2. **Tool-footprint or contact mask** M^{tool} : the visible or predicted contact footprint;
3. **Protected-object mask** M^{obs} : hardware, string anchors, fretboard, or other regions that must not be contacted;
4. **Defect or condition mask** M^{def} : scratch, haze, oxidation, stain, or local loss of gloss when such a region is visually detectable.

Mask-Guided Object-Centric Force-Aware Imitation Learning

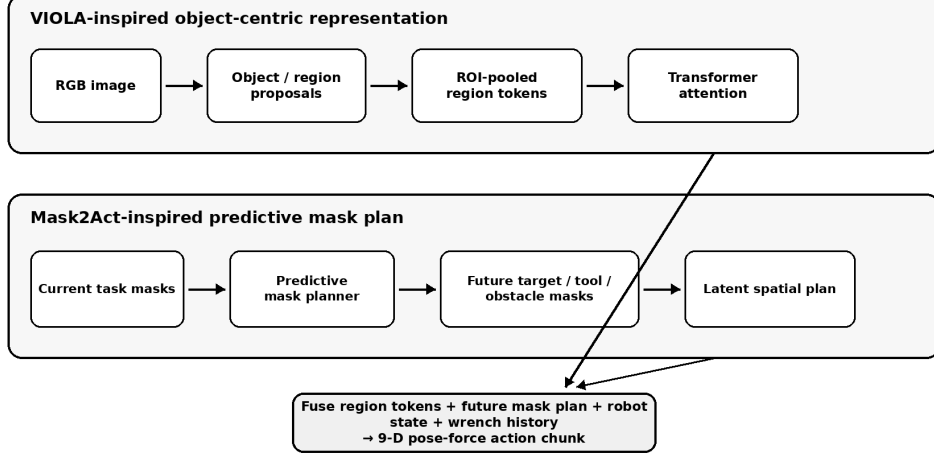


Figure 6.1: Proposed integration of VIOLA-inspired object-centric region tokens and Mask2Act-inspired predictive mask plans with the force-aware IL policy.

Masks can be obtained from manual labels, a task-specific segmentation model, or a pretrained segmentation foundation model followed by temporal refinement. The thesis should distinguish mask-generation error from policy error.

6.3 VIOLA-Inspired Object-Centric Region Tokens

VIOLA uses object proposals from a pretrained vision model to construct object-centric representations and a transformer policy to attend to task-relevant visual factors [17]. The proposed polishing adaptation replaces generic household-object proposals with region proposals relevant to contact: target surface, tool footprint, protected hardware, and defect region.

Given image feature map $\mathbf{Z}_t$, each proposal or mask M_t^i produces a region token through ROI pooling or masked average pooling:

$$\mathbf{r}_t^i = \text{Pool}(\mathbf{Z}_t, M_t^i). \quad (6.1)$$

The set of tokens $\{\mathbf{r}_t^i\}$, a global image token, robot-state tokens, and wrench-history tokens are processed by a transformer. Attention weights provide an interpretable indication of which region contributes to the pose-force prediction, although attention alone should not be treated as proof of causal relevance.

6.4 Mask2Act-Inspired Predictive Mask Plans

Mask2Act learns a planner from action-free video by predicting future masks of task-relevant objects and uses the latent plan to guide an executor trained with robot action data [18]. For polishing, the proposed planner predicts a sequence of future target, tool, and protected-object masks:

$$\hat{\mathbf{M}}_{t+1:t+H} = \mathcal{P}(I_{t-h:t}, \mathbf{M}_{t-h:t}, g), \quad (6.2)$$

where g specifies the local task or target region. The predicted future masks encode how the contact footprint should move over the target while remaining separated from protected hardware. Unlike future RGB prediction, the mask sequence suppresses texture and illumination details that are unrelated to motion.

The action executor conditions on the future mask plan, robot state, and wrench history:

$$\mathbf{A}_t = \mathcal{E}(I_t, \hat{\mathbf{M}}_{t+1:t+H}, \mathbf{s}_t, \mathbf{H}_t^w). \quad (6.3)$$

The future mask sequence is a latent spatial plan, not a substitute for force sensing. It specifies where the task-relevant regions should move in image space, while the wrench history informs whether the physical contact is correct.

6.5 Combined Mask-Guided Force Policy

The full proposed architecture contains two complementary branches. The VIOLA-inspired branch provides object-centric region tokens at the current

time. The Mask2Act-inspired branch predicts future masks as a high-level spatial plan. These representations are fused with robot state and force history before the flow-matching action generator.

The policy can be trained with a combined objective:

$$\mathcal{L} = \mathcal{L}_{\text{action}} + \lambda_{\text{mask}} \mathcal{L}_{\text{future mask}} + \lambda_{\text{cons}} \mathcal{L}_{\text{contact consistency}}, \quad (6.4)$$

where $\mathcal{L}_{\text{action}}$ is the behavior-cloning or flow-matching loss, $\mathcal{L}_{\text{future mask}}$ supervises future masks, and $\mathcal{L}_{\text{contact consistency}}$ encourages agreement between predicted tool masks and measured contact phases. The exact consistency term remains to be finalized.

6.6 Training Strategy

A staged training procedure is proposed:

1. train or adapt the mask extractor on target, tool, hardware, and defect annotations;
2. pretrain the future-mask planner from action-free or demonstration videos;
3. train the force-aware executor using robot demonstrations while freezing or slowly fine-tuning the planner;
4. jointly fine-tune the combined policy with conservative learning rates and validation on unseen lighting and background conditions.

Action-free video is particularly attractive because local polishing demonstrations with synchronized robot action and force are expensive. However, human and robot tool morphology differ, so the predicted mask plan should focus on target and contact-footprint motion rather than exact human hand motion.

6.7 Planned Ablations

The feature-masking study compares:

1. raw RGB baseline;
2. RGB multiplied by a binary target mask;
3. RGB plus mask as an additional channel;
4. VIOLA-inspired object-centric region tokens;
5. Mask2Act-inspired future-mask latent plan;
6. combined region-token and future-mask model;
7. each of the above with and without wrench history.

Evaluation includes task success, local coverage, collision, force metrics, generalization to lighting and background changes, sensitivity to mask noise, and representation analysis using occlusion tests and controlled perturbations. The combined model should be accepted only if the benefit exceeds the added segmentation and training complexity.

Chapter 7

Experimental Design

7.1 Validation Strategy

The experiments are organized from isolated process validation to full-product integration. The sequence is shown in Fig. 7.1.

7.2 Hardware and Test Objects

The target hardware includes a UR10e-class robot, a six-axis force/torque sensor, a polishing spindle, interchangeable broad and local polishing tools, an RGB-D scanner, a local RGB camera, and a rigid guitar fixture. Early tests use coated flat and convex coupons. Geometry experiments use 3-D printed or sacrificial guitar-body replicas before valuable instruments are introduced.

The final object set should contain at least three body classes, for example a double-cut solid body, a single-cut or arched body, and an acoustic or violin-like curved body. Local tests include body edges, cutaways, neck transitions, frets, pickup or bridge regions, and removable metal coupons that reproduce guitar hardware finish.

7.3 Process-Model Calibration

Normal force, sliding velocity, dwell time, tool tilt, spindle speed, pad type, and abrasive compound are varied on coupons. Actual removal is measured using profilometry, thickness change, mass change, or high-resolution surface reconstruction. These tests calibrate the Preston coefficient and the effective contact-footprint model used by Chapters 3 and 4.

Experimental Validation Sequence

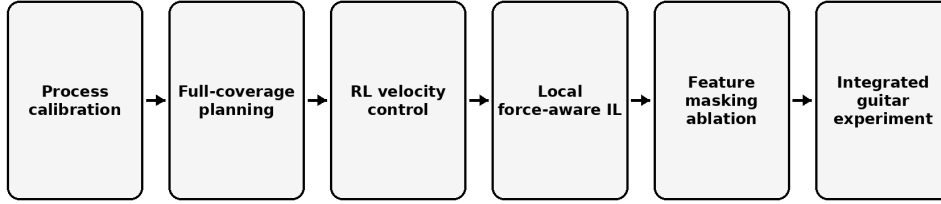


Figure 7.1: Planned experimental validation sequence.

7.4 Full-Coverage Path-Planning Experiments

The path planner is compared with fixed planar raster motion, nominal-CAD projection, constant-spacing mesh projection, curvature-aware spacing, and the proposed tool-footprint-aware planner. Metrics include target coverage, boundary coverage, over-processed overlap, path length, orientation smoothness, manipulator feasibility, collision clearance, and planning time.

7.5 RL Sliding-Velocity Experiments

The same path and admittance controller are used for fixed speed, inverse-Preston control, PID or model-based rate control, and RL velocity modulation. Tests vary curvature, force bias, sensor delay, friction, pad stiffness, and velocity saturation. Both predicted Preston-rate variation and measured-removal variation are reported.

7.6 Force-Aware IL Experiments

Local tasks are evaluated under unseen geometry, initial pose offset, finish color, reflection, wear or oxidation pattern, friction, pad condition, and contact disturbance. Baselines are manual local programming, RGB-only IL, RGB

plus force observation with fixed desired force, and full 9-D pose–force IL. The policy is evaluated on task completion, local coverage, force RMSE, peak force, contact loss, collision, intervention, smoothness, and teaching time.

7.7 Feature-Masking Experiments

Chapter 6 is evaluated separately from the core IL policy. Mask quality is varied systematically through boundary erosion, dilation, random holes, delayed updates, and identity errors. Lighting, background, camera pose, and distractors are changed while geometry remains fixed. The study compares raw RGB, static mask conditioning, VIOLA-style region tokens, Mask2Act-style future masks, and the combined model.

7.8 Integrated Full-Product Experiment

The complete experiment performs CAD or scan registration, full-coverage planning, RL speed modulation on broad surfaces, local IL execution at designated regions, and final quality inspection. The integrated framework is compared with geometry-only automation and a semi-automated framework in which a human manually specifies the local paths.

7.9 Baselines

7.10 Evaluation Metrics

7.11 Statistical Analysis

Each method-condition pair is repeated across multiple specimens and trials. Method and test condition are treated as fixed factors; specimen and guitar model can be treated as random effects in a mixed-effects analysis.

Table 7.1: Planned experimental baselines.

ID	Method	Purpose
B0	Manual expert	Human quality and time reference
B1	Basic CAD coverage + fixed speed	Conventional geometric baseline
B2	Advanced region/contact-point planning	Strong geometry baseline
B3	B2 + manually programmed local paths	Semi-automated baseline
B4	RGB-only IL, 6-D pose, fixed force	Vision-only learning
B5	RGB + wrench observation, fixed force	Force observation only
B6	9-D pose-force IL	Core local policy
B7	Fixed path speed	Process baseline
B8	Inverse-Preston speed control	Analytical strong baseline
B9	PID/model-based rate control	Feedback strong baseline
B10	RL velocity-scale policy	Proposed process policy
B11	VIOLA-style region tokens	Object-centric masking
B12	Mask2Act-style future masks	Predictive latent plan
Full	Global planner + RL + IL + masking	Integrated framework

Table 7.2: Evaluation metrics.

Category	Metrics
Coverage	Coverage ratio, uncovered area, boundary completion, excessive overlap
Removal	Preston-rate CV, cumulative-removal CV, target-band violation, measured removal
Surface quality	Roughness, gloss uniformity, residual defect, edge damage
Contact	Force RMSE, peak force, dF/dt , contact loss, tool-normal error
Safety	Collision, near-collision, overload, retraction, intervention
Efficiency	Scan, planning, teaching, cycle, and non-processing time
Generalization	Geometry, pose, appearance, wear, friction, and pad-condition success

Failure modes are categorized as reconstruction error, coverage-planning error, kinematic infeasibility, local-policy perception error, contact instability, termination error, mask error, and out-of-distribution behavior.

7.12 Planned and Existing Results

Only internal simulation results for RL adaptive velocity are currently available. All other numerical results remain planned experiments and must be reported without fabrication. The final thesis should distinguish simulation, coupon, replica, and real-guitar results clearly.

Chapter 8

Conclusions and Future Work

8.1 Conclusions

This thesis draft defines a hybrid framework for full-product robotic surface finishing. The central design decision is to avoid using one method for every region and decision level. Tool-footprint-aware geometric planning provides explicit full coverage of broad CAD surfaces. Reinforcement learning optimizes only the sliding velocity along the reference path, preserving geometric correctness and low-level safety. Force-aware imitation learning transfers local pose–force strategies to edges and contact-sensitive regions. Feature masking applies object-centric proposals and predictive future masks to reduce the influence of reflections and background distractors.

The methods are connected through a common 9-D pose–force trajectory and a deterministic admittance controller. This interface enables fair comparisons and modular development. The scientific contribution will depend on experimental evidence that the integrated system improves coverage, removal uniformity, local contact behavior, programming time, and generalization relative to strong geometric, analytical, and learning-based baselines.

8.2 Limitations

The tool-footprint model is initially approximate and depends on pad compression, curvature, and force. The RL policy may fail to outperform an analytical inverse-Preston controller in well-modeled conditions. The IL policy depends on demonstration coverage and can fail on unseen hardware or

wear states. Mask extraction can introduce an additional failure mode, and predictive masks may be visually correct but physically inconsistent with force and depth. A guitar demonstrates the target method-selection problem but does not prove universal industrial applicability.

8.3 Future Work

Future work includes automatic assignment of global and local regions using reconstruction confidence and planner feasibility; force/moment-based contact-point estimation; joint optimization of force, speed, spindle speed, and admittance gains; nonlinear and online material-removal modeling; sim-to-real RL transfer; tool changing between broad and local pads; active camera positioning; quality-aware termination; and validation on furniture, molds, consumer-product housings, violins, and automotive trim.

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Appendix A

Implementation and Experiment Checklists

A.1 Current Implementation Status

- CAD/mesh path generation and dense pose–force trajectory conversion: partially implemented.
- Low-level admittance controller: available in the existing robot-control stack.
- RL velocity-scale policy: simulation prototype and preliminary internal results available.
- Force-aware IL with direct 9-D output: research architecture defined; final implementation and real-robot validation pending.
- VIOLA- and Mask2Act-inspired feature masking: conceptual design; implementation pending.

A.2 Recommended Dataset Fields

Each demonstration episode should contain timestamps, RGB images, masks, end-effector position and orientation, linear and angular velocity, six-axis wrench, desired force, spindle state, action chunk, region identifier, phase labels, intervention labels, and success or failure annotations.

A.3 Build Notes

The project is compiled from `ThesisMain.tex`. A typical command is `latexmk -pdf ThesisMain.tex`. The cover can be compiled separately from

`ThesisCover.tex`. The author, advisor, submission date, final figures, and experimental results are placeholders that must be updated before submission.

국문요약

기하학적 경로계획, 강화학습, 힘 인지 모방학습 및 특징 마스크 기반 하이브리드 로봇 표면 마감 프레임워크 개발

본 학위논문 초안은 하나의 제품 전체를 자동으로 폴리싱하기 위한 하이브리드 로봇 표면 마감 프레임워크를 제안한다. 제안 프레임워크는 CAD 또는 재구성된 표면 전체를 커버하는 공구 폭 기반 전역 경로계획, 실제 접촉력에 따라 sliding velocity를 조절하는 강화학습, 에지 및 국소 난구간에서 위치·자세·목표 힘 궤적을 생성하는 힘 인지 모방학습, 그리고 반사광과 배경 정보에 대한 의존을 줄이는 특징 마스크로 구성된다.

전역 경로계획에서는 TCP 자체가 아닌 실제 polishing pad의 유효 접촉 폭과 overlap ratio를 이용해 경로 간격을 결정하고, 공구가 이동하며 형성하는 swept footprint의 합집합이 목표 표면을 포함하도록 full-coverage path를 생성한다. 강화학습 정책은 로봇 관절이나 6차원 자세를 직접 출력하지 않고, 고정된 기준 경로의 진행 속도 scale만 조절하여 Preston equation 기반 가공률 편차를 줄인다. 힘 인지 모방학습 정책은 RGB 영상, 로봇 상태 및 6축 힘·토크 이력을 입력으로 받아 위치 3축, 자세 3축, 목표 힘 3축으로 이루어진 9차원 action chunk를 생성하고, 저수준 admittance controller가 이를 추종한다.

특징 마스크 장에서는 VIOLA의 object proposal 기반 object-centric representation과 Mask2Act의 future object-mask latent plan을 현재 IL framework에 적용하는 방법을 제안한다. 대상 surface, tool footprint, 보호해야 할 hardware 및 결함 영역을 mask로 표현하고, object-centric region token과 미래 mask sequence를 힘 정보 및 로봇 상태와 융합해 9차원 pose-force action을 생성한다. 이 부분은 아직 구현 전의 설계안으로서 정적 mask conditioning, VIOLA-inspired region token, Mask2Act-inspired future-mask plan 및 결함 모델을 비교하는 실험을 계획한다.

실증 대상물은 넓은 도장 곡면과 에지, 컷어웨이, 프렛, 픽업, 브릿지 등 다양한 국소 금속 구간이 하나의 제품에 공존하는 기타로 설정한다. 최종 연구는 전체 커버리지, 가공량 균일도, 접촉력 안정성, 안전성, 교시 및 프로그래밍

효율과 다양한 형상 및 사용 상태에 대한 일반화 성능을 종합적으로 평가한다.

주요어: 로봇 표면 마감, 전면 커버리지 경로계획, 공구 접촉 폭, 강화학습, 힘 인지 모방학습, 특징 마스킹, 어드미턴스 제어